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Task Description:	Protein crystallization experiments in microgravity have often generated unexpectedly low or high numbers of crystals; both outcomes may doom experiments designed to obtain a small number of well-separated crystals for x-ray structure studies. This unpredictability has been associated with the suppression of buoyancy-driven convection, as supported by data indicating that protein crystal nucleation rates reach a maximum at an optimal convection velocity. To elucidate the effects of solution convection on crystal nucleation--a crucial issue in several fields of science and engineering--we propose to test two mechanisms: 1. Convection enhances or suppresses the formation of the dense liquid clusters, within which crystal nuclei form. 2. Convection enhances or suppresses the formation of crystal nuclei within the clusters. Tests of mechanism 1. require characterization of the dense liquid clusters at different rates of convection, including its complete absence, only possible in microgravity. We propose to use the Light Microscopy Module (LMM) aboard the International Space Station (ISS) and two variants of the Advanced Colloid Experiment (ACE): ACE-M and ACE-C. To prepare for microgravity experiments, we will design microscopy techniques that can then be implemented on the LMM. We will extend a newly-developed method, differential dynamic microscopy, to quantify the dynamics of the dense liquid clusters. Its variation with solution composition and the rate of convection will be used to identify the cluster formation mechanisms. Towards tests of mechanism 2, we will determine the supersaturation dependencies of the crystal nucleation rates in both quiescent and flowing solutions. For this, we will implement novel microfluidics devices made of SU-8 and glass, which allow long-duration experiments under steady conditions. Finally, we will use the insight on convection effects on nucleation to establish procedures leading to higher perfection of protein crystals and test them with proteins of interest to biomedical sciences.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Solution crystallization underlies a broad range of industrial, laboratory, and physiological processes. Single solution-grown crystals of inorganic salts or mixed organic-inorganic materials are used in non-linear optics elements and for other electronic and optical-electronic applications; chemical products and production intermediates are precipitated as crystals in thousands-of-tons amounts. Another area which relies on solution-grown crystals is pharmacy: the slow crystal dissolution rate is used to achieve sustained release of medications: small-molecules organic, or proteins such as insulin, interferon-alpha, or the human growth hormone. If the administered dose consists of a few equidimensional crystallites, steady medication release rates can be maintained for longer periods than for doses comprised of many smaller crystallites. The formation of protein crystals and crystal-like ordered aggregates underlies several human pathological conditions. An example is the crystallization of hemoglobin C and the polymerization of hemoglobin S that cause, respectively, the CC and sickle cell diseases. The formation of crystals in the eye lens underlies the pathology of several forms of cataract. A unique example of benign protein crystallization in humans and other mammals is the formation of rhombohedral crystals of insulin in the islets of Langerhans in the pancreas. Traditionally, protein crystals have been used for the determination of the atomic structure of protein molecules by x-ray crystallography; this method contributes ~87% of all protein structures solved, with the majority of the other determinations carried out by Nuclear Magnetic Resonance (NMR) spectroscopy. Due to its crucial place at the start of the crystallization process, the nucleation of crystals determines many properties of the emerging crystalline phase. It is obvious that the nucleation selects the polymorphic form and if a different polymorph is desired, conditions at which its nucleation is faster than that of the other possible polymorphs should be sought. If nucleation is fast, many crystals form nearly simultaneously. Their growth depletes the medium of solute and may lead to cessation of nucleation at the later stages of crystallization. Thus, the majority of crystals grow to approximately identical sizes. In contrast, if nucleation is slow and fewer crystals nucleate at a time, the supersaturation in the solution drops slowly, the nucleation of new crystals continues and a population of crystals of various sizes forms. Ultimately, if nucleation is hindered everywhere in the growth container but at a few selected spots, crystals only nucleate at these spots and grow large before the solution is depleted of nutrient. Hence, control of nucleation is a means to control size, size distribution, polymorphism and other properties of the crystals.
	Towards these goals, during the reported period, the Vekilov laboratory worked on five research tasks: 1. Improvements of the differential dynamic microscopy (DDM) method to characterize the protein rich clusters in protein solutions compatible with the light microscopy module (LMM) aboard ISS. 2. Characterization of the protein rich clusters in solutions of the protein RNase-A. 3. Identification of the transient protein dimer that underlies the formation of the protein rich clusters. 4. Characterization of the diffusive dynamics of particles with time-dependent asymmetric microscopy intensity profiles 1. To employ a light microscope for the characterization of the protein rich clusters, we have modified and improved a recent method to study used to quantify dynamics of sub-micrometer particles in solutions from fluctuations in intensity in optical micrographs, differential dynamic microscopy (DDM). Biological objects often scatter light weakly and are frequently smaller than the diffraction limit, complicating measurements of their dynamics. Differential dynamic microscopy is well established for monodispersed particles but has not been applied to polydispersed biological nanoparticles. We used DDM to measure dynamics of protein-rich liquid clusters, which are polydispersed nanoscale objects, in protein solutions, whose size ranged from tens to hundreds of nanometers and whose total volume fraction was less than 10-5. For solutions of two proteins, lysozyme and hemoglobin A, we measured the dynamics of clusters using DDM and evaluated their diffusion coefficients from the dependence of the diffusion lag time on the scattering wave vector. In the most recent funding year, we tested major advantage of brightfield DDM (b-DDM) to resolve the separate dynamics of each component of a bidisperse suspension of 50 nm and 1 μm nanoparticles in low ranges of wave vectors (0.88 – 3 μm-1) while common scattering methods including DLS (dynamic light scattering) would fail to capture the distinct diffusional behavior of each nanoparticle. We tested three different suspensions at volume fraction ratio of large to small particles varying from 0.003, 0.01 to 0.03 by keeping the volume fraction of the small particles at phi(s)=10-3. Following our expectation, for volume fraction ratio of 0.03 where DLS resolve the large particles at q = 6.84 μm-1, DDM also resolves separate diffusion relaxation time of each particle population. Interestingly, we observe that due to

Task Progress:

high qR values for large particles (i.e., q.R > 1) the scattered light turns out to be incoherent which affects the measured dynamics more specifically for 1 < q.R < 2 at which the interference of transmitted and scattered light turns out to be deconstructive. In order to probe this effect, we fit all the data with incoherent model and we capture certain ranges of wave vectors for which incoherency time is comparable to the relaxation times and accordingly, we corrected our relaxation times (see Vekilov et al., 2016 in Bibliography section).

We also showed that for ratios 0.01 and 0.003 where DLS fails to capture dynamics of large particles even at the lowest accessible wave vector q= 6.84 µm-1, DDM is still able to resolve each particle population using our global two exponent fits where the relaxation time ratio of large to small particles are globally fitted at all wave vectors and large particles relaxation time, incoherency time, amplitude and noise parameters are fitted separately for each individual wave vectors.

A paper on this improvement of the DDM method has been published, and another one is in preparation.

2. In experiments here with the acidic protein Ribonuclease A (RNase A), we studied the effect of ionic strength, urea, and temperature on the behaviors of the mesoscopic protein rich clusters. We employed Brownian microscopy (BM), static light scattering (SLS), and several biochemical assays. The cluster size distribution was measured for three days with BM and revealed a consistently narrow size distribution through this time period, indicating stability of clusters. The clusters respond to the change in protein concentration; this change is exhibited as decrease in cluster volume fraction as the protein concentration was increased indicating these clusters are not irreversible aggregates. The intermolecular forces were varied by changing the ionic strength of solution with sodium chloride, ammonium sulfate, and sodium acetate and quantified by measuring second virial coefficient by SLS. The data revealed that strong intermolecular repulsion decreases the cluster size and cluster volume fraction. The hydrophobic interactions between the exposed hydrophobic residues due to partial unfolding and solvent was studied by the addition of urea, a chaotropic agent that destroys the shell of structured water around protein molecules. Elman's assay was used to quantify the free thiols and the degree of unfolding was studied using ANS and ThT assays. The unfolding of protein by breaking disulphide bonds by heating showed an increase in cluster size and cluster volume fraction indicating the role of protein unfolding to aid the cluster formation. These experiments demonstrated the relevance of partial unfolding required in formation of clusters and the effect of urea as a chaotropic agent.

3. According to the two-step nucleation mechanism, crystal nuclei form within pre-existing dense liquid clusters that then order into a nucleus. This mechanism separates the transition from liquid to crystal across two order parameters, structure and density. Clusters with radii of approximately 100 nm, which capture from 10-7 to 10-3 of the total protein, have been observed with many proteins using sensitive methods such as dynamic light scattering, and moreover have been shown to host crystal nucleation. Neither the size of clusters nor their low volume fraction can be explained using existing models. In solutions of single-chain proteins, a new theory proposed to explain these features suggests that partially misfolded proteins form weak dimers, which diffuse into and out of the clusters. Such dimers were predicted to be non-covalent and to capture no more than 10-3 of the total protein. Experimental confirmation of the predicted dimers, however, has proven elusive, in large part because the dimers are dilute, transient, and chemically nearly identical to the monomers.

In the most recently funded year, we developed a method to detect these highly dilute dimers in isotopically resolved mass spectra using Fourier transform analysis. Application of this method to simulated mass spectra reproduces trends predicted by analysis of simple model functions. Using this method, we have detected non-covalent dimers in 40 mg/mL lysozyme solutions, the first direct confirmation of the existence of dimers implied by the proposed theory for the two-step mechanism.

4. Over the years several computer algorithms for particle tracking from a sequence of microscopic images have been developed. On each image, particles are identified and the coordinates of their centers are determined, resulting in a time series of positions. General tracking methods deal with cases in which the intensity profile of a particle is radially symmetric. As such the center is allocated to the point of maximum intensity or the intensity centroid. More sophisticated techniques use a variety of fitting algorithms to provide sub-pixel resolution of the particle coordinates. As the peak of the point spread function of a diffraction-limited spot can, in most cases, be approximated by a Gaussian function, many fitting algorithms use Gaussian profiles. The radial centering technique uses the direction of the gradient in a radially symmetric intensity pattern to identify the intensity center. This technique does not employ iterative nonlinear fitting and is faster than fitting algorithms.

Several recently studied systems appear to challenge even advanced particle tracking algorithms. For instance, in images of micron-size liquid droplets the recorded intensity patterns are asymmetric and fluctuate with a characteristic time comparable to their diffusion time. Another example comes from particle tracking by oblique illumination microscopy, a method that records the intensity scattered from submicron and micron-size particles in which the identification of the particle center is hampered by the asymmetry of the scattered intensity pattern. To quantify the diffusive motion for such particles and droplets, here we put forth a fast and easy method to implement SPT algorithm. We employ a local spatial cross-correlation function to identify the displacement of a particle between two frames. We implement radial centering of the computed cross-correlation functions to evaluate the travelled distance with sub-pixel resolution. In its current implementation, the method is only applicable to relatively dilute solutions since it requires non-overlapping trajectories of the tracked particles. Importantly, the proposed algorithm does not rely on identification of the particle center in any single image, but reconstructs the particle trajectory from the displacements between pairs of images.

We demonstrate that particle tracking with this method enables an accurate characterization of the dynamic behavior of single particles. We employ the proposed algorithms to obtain trajectories of submicron and micron-size particles and droplets of protein dense liquid. From the recorded trajectories, we evaluate the diffusion coefficient of spherical latex particles of known size, using least squares fits of the mean square displacement as a function of time, and compare its value to the diffusivity determined by dynamic light scattering. We use the correspondence between the two values as a quantitative indicator of the performance of the method. Applying this criterion, we demonstrate that the proposed cross-correlation method yields a more accurate estimate of the particle diffusivity than several common tracking methods.

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